



NICOLA CAMPANELLI

Contact

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LinkedIn
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Skills & Tools

Software & Programming
Matlab, Python, Git, LaTeX

Simulation & Modeling
Matlab Simulink, GT-Suite

Technical
Data analysis, model design and validation, control optimization

Office Productivity
Microsoft Word, PowerPoint, Excel

Soft Skills
Technical presentation, teamwork

Languages

Italian **C2**
Native speaker

English **C2**
Proficient

German **A2**
Basic

Motivated Engineer with international technical experience in vehicle simulation, control optimization, and energy management systems. Skilled in data analysis, model design and validation, and performance optimization of complex systems. Eager to apply technical knowledge across diverse domains, with a strong drive for continuous learning and innovation. Seeking an engineering role focused on the study, design, development, or control of sustainable and smart solutions within any technical field. Recognized as a reliable task finisher and a disciplined, fast learner who thrives in collaborative settings.

Professional Experience

- Jan 2026 – Present

Simulation Engineer
GammaTech Engineering – Torino, Italy
 - Perform vehicle longitudinal dynamic simulations with driver-in-the-loop on a Matlab Simulink / GT-Suite model of a high-performance Hybrid Electric Vehicle, supporting the full process of design and development of the vehicle.
 - Analyze complete powertrain behavior and perform vehicle-level energy analysis across standard homologation cycles and custom maneuvers.
 - Design test maneuvers, run simulation campaigns, and post-process results in order to deliver results through technical reports and presentations.
- Oct 2024 – Jul 2025

Research Fellow
Center for Automotive Research – Columbus, OH, USA
 - Conducted Techno-Economic Analysis of Long-Duration Battery Energy Storage Systems (BESS) to support large-scale deployment, focusing on performance, profitability, and scalability.
 - Developed a simulation model of the Distributed Energy Resources (DERs) plant on Matlab, integrating an optimization control algorithm for the BESS operation.
 - Delivered results about the financial viability of the BESS, pointing out critical insights about its early adoption.
- Nov 2023 – Oct 2024

Research Fellow
Polito E3 – Torino, Italy
 - Development of a Reinforcement Learning (RL) based Energy Management System (EMS) for a Hybrid Electric Vehicle.
 - Development of a simulation platform in the Python software environment for Hybrid Electric Vehicles with a focus on energy management strategies.
 - Development of a Neural Network framework to model a physical system (Vehicle HVAC), obtaining results very close to physics-based models but with much lower computational time.

Education

- 2021 – 2023

Master’s Degree in Automotive Engineering
Politecnico di Torino – Torino, Italy
- 2017 – 2020

Bachelor’s Degree in Automotive Engineering
Università degli Studi di Modena e Reggio Emilia – Modena, Italy

Publications

- Journal paper

“Development of a Soft-Actor Critic Reinforcement Learning Algorithm for the Energy Management of a Hybrid Electric Vehicle”
DOI: 10.4271/2024-37-0011
- Journal paper

“Optimal control and techno-economic analysis of long-duration battery energy storage system applications”
DOI: 10.1016/j.est.2026.121863